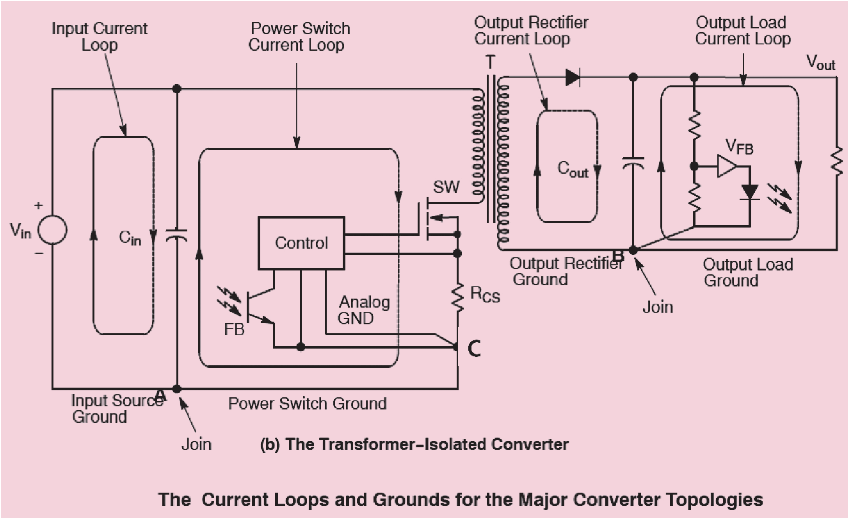

Fig. 4(b): Converter circuit (Source: SMPSRM/D, Rev 4, Apr 2014)

that enough decoupling capacitance is present to filter away the high-frequency noise.

6. Use high-frequency, low-impedance capacitors at both input and output of a transformer, combined with other types of filters and capacitors. Also, minimise the loop area of fast di/dt current paths, which can be done using toroidal-core inductor for output filters or by avoiding rod-shaped inductors that generate an H-field because of their open core shape.

7. Pay attention to the reverse recovery catch diode characteristics, which can be a source of H-field emissions. Forward recovery, if too slow, will delay the transition time and increase the E-field emission. Although Schottky diodes do not have reverse-recovery issues, they have a parasitic capacitance that resonates with the parasitic inductance of the other components and PCB. If necessary, place an RC snubber across the catch diodes.

By following these steps, you will greatly reduce EMI/EMC in an SMPS.

Efficient PCB layout

If the base is not correct, then no circuit will work in the system. So, setting the base, that is, the correct PCB design, is highly important. A perfect SMPS cannot be made without a perfect PCB.

From Fig. 4(a) we can see that there are four main current loops. A and B are AC current loops with high di/dt. The other two are input and output current loops, which are at low frequency. Loops A and B should be laid first because they are high-frequency loops. Their tracks have to be wide and as small as possible, minimising the resistive and inductive effects on the components. The other two low-requency loops should be connected directly to their filter capacitor terminals; otherwise switching noise can bypass the filtering action and create conducting interference.

The word 'Join' mentioned at the bottom of the diagram refers to the points where ground tracks should meet both high and low-frequency sections (capacitor negative terminals), which is generally missed and leads to a lot of filtering and EMI issues. This can be implemented for both isolated and non-isolated circuits.

Another example is the transformer-isolated circuit to which the same rule applies here, that is, the same precautions need to be taken for isolated circuit components as discussed for non-isolated circuits.

Another important factor in PCB layout is ground. It is well known that ground is the main culprit behind every noise. However, it is also required as it forms the reference connection to the entire supply. Each ground has its own set of signals and can create a major problem if not connected properly.

In addition to the four ground loops that were previously discussed, there is a fifth ground loop, which is a low-level analogue control ground or controller ground, critical for proper operation of supply.

The word 'Join' (at the bottom-left in Fig. 4) indicates that the grounds, which form the major current loop, must be connected. Here, the connecting point between high-level AC ground, and input and output ground is the negative terminal of the filter capacitor to avoid EMI/EMC issues.

Next, the analogue control ground must be connected to the point where IC and its associated measuring circuit meet (indicated by C in Fig. 4(a) and 4(b)). If grounded above C, then any AC noise at large signals will sum up and can disturb the measuring of the control circuitry, which will impact the functioning of the overall system. Thus, the purpose of connecting the control ground at the lower side of the current sensing resistor or output voltage resistor divider is to form a Kelvin contact where any common-mode noise is not sensed by the control circuitry and does not impact the 'brain' of the SMPS. You need to sense properly for controlling the circuit.

So, these measurements need to be clear, and for that the grounding has to be very clean near the controller. PCB design is a critical factor in any design and should be done in a very focused manner to keep your designs running smoothly for many years.

Top mistakes, which designers make when designing SMPS:

1. Not doing enough design simulations, which help a lot in designing circuits efficiently.

2. Not doing proper calculations. (If you ignore them, you will face problems later in the production.)

3. 'Overdesigning' SMPS for it to run longer. If you do correct calculations, you can derive the best value of each component.

4. Improper testing. EFY

The article is based on the talk 'Designing Efficient SMPS: Core Of Everything Electronics' by Nupur Jindal, a Power Electronics professional, which was presented at the March edition of Tech World Congress 2021. The article is compiled by Vinay Prabhakar Minj, a technology journalist at EFY